

Cervical Shortening 04. Prior Spontaneous Preterm Birth

Companion reading handout for simultaneous listening.

Audio course on cervical shortening during pregnancy. Episode four. Short cervix with prior spontaneous preterm birth.

Opening frame

In the last episode, we managed the clean low-risk singleton: no prior spontaneous preterm birth, asymptomatic, closed cervix, short cervix before twenty four weeks.

This episode changes one fact.

The patient has a prior spontaneous preterm birth.

That single fact changes the meaning of the same cervical length.

Dominant model

The dominant model is this.

History changes the pretest probability that a short cervix is clinically meaningful.

A twenty three millimeter cervix is not the same finding in every patient. In a no-prior-spontaneous-preterm-birth singleton, it is a risk marker in a lower baseline risk population. In a patient with a prior spontaneous preterm birth, it may be a sign of recurrence biology. The number is the same. The patient is not.

So the first task is to reconstruct the prior pregnancy.

Do not accept the phrase, she delivered early, as a complete history.

Ask: what gestational age? Was it spontaneous labor, PPRM, painless cervical dilation, bleeding, abruption, fetal anomaly, infection, indicated delivery, severe preeclampsia, or fetal growth restriction? Was there a cerclage? Was there cervical surgery? Was there a midtrimester loss? Were membranes exposed? Was there a neonatal survival issue? Was the delivery before thirty four weeks, before twenty eight, or periviable?

Why does this matter?

Because a prior spontaneous preterm birth after contractions is not identical to recurrent painless midtrimester dilation. PPRM is not identical to indicated preterm birth for preeclampsia. A prior loop excision plus painless dilation is not the same as

spontaneous labor at thirty five weeks. The recurrence pathway determines how you interpret the current cervix.

Now the surveillance logic.

Patients with prior spontaneous preterm birth are often followed with serial transvaginal cervical length surveillance in the midtrimester, depending on local protocol. ACOG's updated progesterone advisory emphasizes serial endovaginal cervical length measurement in singleton pregnancy with prior spontaneous preterm birth. The exact interval can vary, but the principle is that the cervix is watched because history raises baseline risk.

Now the treatment threshold.

When prior spontaneous preterm birth and current short cervix meet, cerclage becomes a real discussion.

Owen two thousand nine is central. Women with prior spontaneous preterm birth before thirty four weeks were screened, and those whose cervix shortened to less than twenty five millimeters were randomized to cerclage or no cerclage. The overall primary outcome, birth before thirty five weeks, was lower with cerclage but did not reach the conventional significance threshold for the whole group. But important signals emerged: previable birth and perinatal mortality were lower, and the benefit appeared concentrated in the very short cervix subgroup, especially less than fifteen millimeters.

The teaching point is not, cerclage for everyone with a prior preterm birth.

The teaching point is: prior early spontaneous preterm birth plus a short cervix identifies a subgroup where cerclage can matter, especially when the cervix is very short.

Now compare guideline language.

NICE is explicit. If a woman has both a history of spontaneous preterm birth or pregnancy loss from sixteen plus zero to thirty four plus zero weeks and a transvaginal cervical length of twenty five millimeters or less between sixteen plus zero and twenty four plus zero weeks, offer a choice of prophylactic vaginal progesterone or prophylactic cervical cerclage after discussing risks and benefits.

NICE also says to consider prophylactic vaginal progesterone when only one of those risk factors is present: either prior spontaneous preterm birth or midtrimester loss, or a short cervix. And it says to consider prophylactic cerclage when there is a short cervix and a history of PPROM.

FIGO uses a cerclage taxonomy. History-indicated cerclage should be offered to women with three or more prior preterm deliveries or midtrimester losses.

Ultrasound-indicated cerclage should be offered to women with cervical length below twenty five millimeters when they have one or more spontaneous preterm births or

midtrimester losses. FIGO also says there is no clear evidence of benefit from cerclage in high-risk women without prior preterm birth or midtrimester loss.

ACOG's progestogen advisory adds another important clarification. Vaginal progesterone may be considered in a patient with singleton pregnancy, prior spontaneous preterm birth, and a shortened cervix. But vaginal progesterone has not been proven effective for recurrent preterm birth prevention when the cervix is not shortened. And intramuscular 17-OHPC is not recommended for primary prevention of recurrent spontaneous preterm birth.

So how should the resident think?

Do not turn guideline differences into confusion. Turn them into phenotype-based reasoning.

If the patient has prior spontaneous preterm birth and the cervix is not shortened, the conversation is different from the patient with prior spontaneous preterm birth and cervical length eighteen millimeters.

If the cervix is twenty four millimeters, the decision may be preference-sensitive and local-protocol-sensitive.

If the cervix is fourteen millimeters with prior early spontaneous preterm birth, the cerclage signal is stronger.

If the cervix is dilated, you are no longer just doing ultrasound-indicated cerclage. You are entering physical-exam-indicated cerclage logic.

Definitions and categories

Let us define the cerclage categories because they prevent sloppy thinking.

History-indicated cerclage is placed because the past obstetric history is compelling, often recurrent painless midtrimester losses or prior exam-indicated cerclage. It is usually placed prophylactically in the late first or early second trimester after confirming viability, depending on local protocol.

Ultrasound-indicated cerclage is placed because a high-risk patient, usually with prior spontaneous preterm birth, develops a short cervix on surveillance.

Physical-exam-indicated cerclage is considered because the current pregnancy has painless cervical dilation, often with exposed membranes, before viability or periviability, after excluding labor, PPROM, infection, bleeding, and major fetal problems.

These are not interchangeable. A patient can move from one category to another, but the indication must be named.

Now a case.

Clinical cases

A patient is eighteen weeks. Singleton. Prior spontaneous preterm birth at thirty one weeks after spontaneous labor. Current TV cervical length thirty two millimeters.

The resident should not place cerclage because of history alone unless the history suggests classic recurrent cervical insufficiency or local protocol has another indication. Continue surveillance. Discuss current progestogen guidance carefully. Do not use 17-OHPC.

Second case.

Twenty one weeks. Prior spontaneous preterm birth at thirty weeks. Current TV cervical length twenty two millimeters.

Now the cervix is short. This is a real discussion. Depending on guideline system and local practice, prophylactic vaginal progesterone and ultrasound-indicated cerclage are both part of the conversation. NICE would offer a choice if the prior event fits the sixteen to thirty four week range and the cervix is twenty five millimeters or less. FIGO supports ultrasound-indicated cerclage in prior spontaneous preterm birth or midtrimester loss with cervical length below twenty five. The resident should involve MFM level decision-making rather than reduce this to, progesterone for everyone.

Third case.

Twenty one weeks. Prior spontaneous preterm birth at twenty eight weeks. Cervical length twelve millimeters.

This is where the cerclage signal becomes stronger. Owen's benefit was concentrated in the very short cervix subgroup. The resident should assess symptoms, exclude PPROM, bleeding, and infection, evaluate for dilation, and discuss ultrasound-indicated cerclage strongly in the appropriate setting.

Fourth case.

Twenty one weeks. Prior painless dilation at twenty weeks with pregnancy loss. Current cervix is not yet short.

This may be a history-indicated cerclage phenotype, not a wait-for-shortening phenotype. The history matters more than the current measurement alone.

Now let us talk about progesterone in this group.

Vaginal progesterone is still biologically plausible when the cervix is short. ACOG says it may be considered for prior spontaneous preterm birth with a shortened cervix. But the resident must not say, prior preterm birth equals progesterone, and stop there. The post-17-OHPC era is more nuanced. Cervical length matters. The treatment target matters. Local protocols matter.

Do not confuse vaginal progesterone for short cervix with old intramuscular 17-OHPC practice.

Now, what about adding progesterone after cerclage?

Evidence is limited. Some retrospective data suggest possible benefit, but SMFM states evidence is insufficient for a definitive recommendation. This is a perfect example of how to teach uncertainty. Say what is known. Say what is not known. Do not pretend an unresolved adjunct question is settled.

Patient counseling

Now patient counseling.

For prior spontaneous preterm birth plus short cervix, the patient often carries trauma from the last pregnancy. She may want the most aggressive treatment. Or she may fear procedures. The resident needs a clear explanation.

Try this:

Your prior preterm birth changes how we interpret this cervical length. In someone without that history, we often treat a short closed cervix with vaginal progesterone alone. With your history, especially because the cervix is now shorter than twenty five millimeters, cerclage becomes a real option to discuss. The decision depends on how early the prior birth was, whether it was labor or PPROM or painless dilation, how short the cervix is now, whether the cervix is dilated, and whether there are signs of infection, bleeding, contractions, or ruptured membranes.

That counseling does not hide complexity. It organizes it.

Final synthesis

Let us close with the resident reflex.

In prior spontaneous preterm birth, do not manage the cervix as if the history were absent.

Reconstruct the prior pregnancy. Use standardized TV ultrasound. If the cervix is below twenty five millimeters before twenty four weeks, think seriously about ultrasound-indicated cerclage and vaginal progesterone strategy, especially when the cervix is very short. If there is dilation, change categories. And do not use 17-OHPC as a shortcut from older practice.

The same cervical length means more when the past pregnancy has already shown you the pathway.